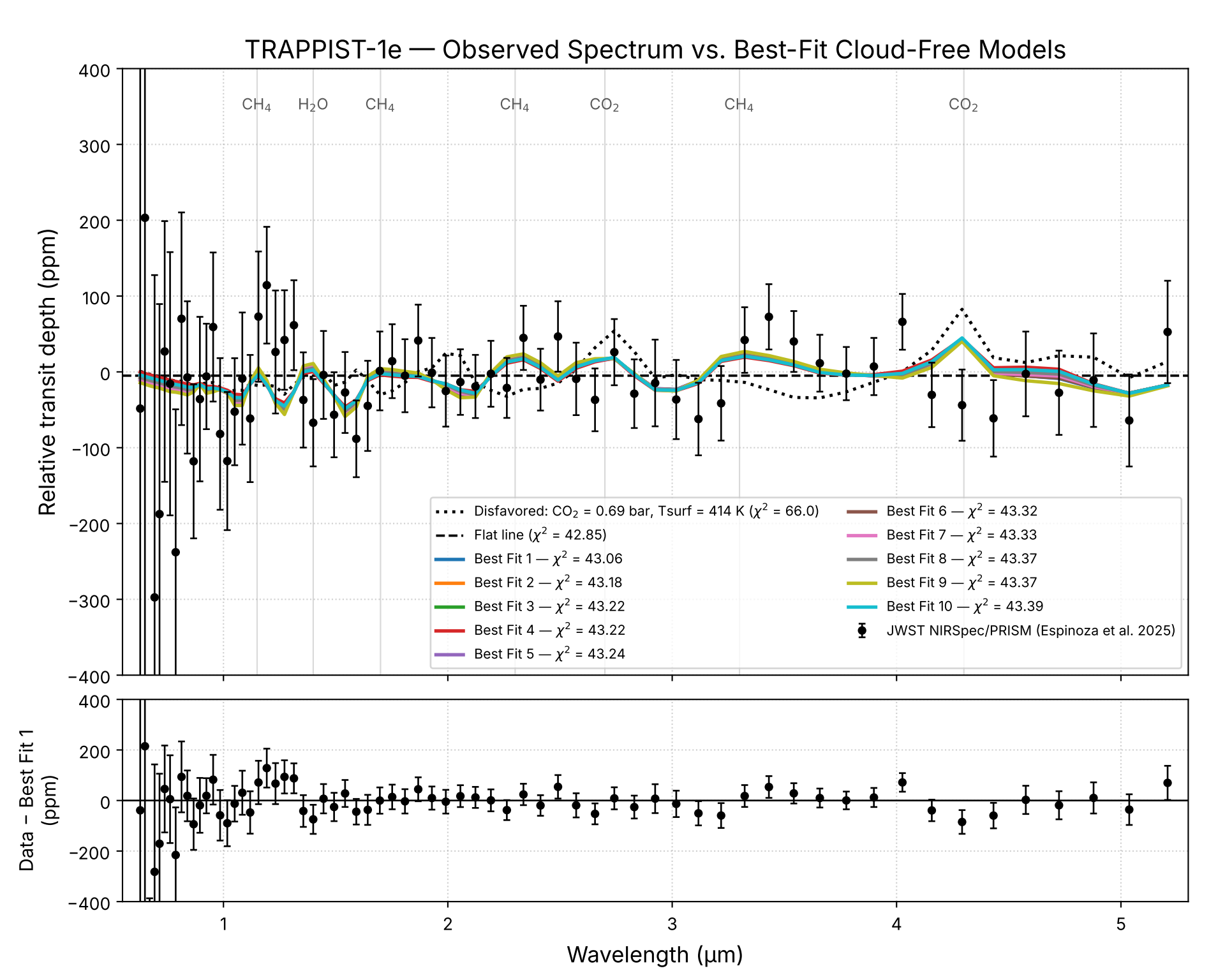
TRAPPIST-1e — Observed Spectrum vs. Best-Fit Cloud-Free Models
CH4
H2O
CH4
CH4
CO2
CH4
CO2
Relative transit depth (ppm)
Disfavored: CO2 = 0.69 bar, Tsurf = 414 K (χ2 = 66.0)
Flat line (χ2 = 42.85)
Best Fit 1 — χ2 = 43.06
Best Fit 2 — χ2 = 43.18
Best Fit 3 — χ2 = 43.22
Best Fit 4 — χ2 = 43.22
Best Fit 5 — χ2 = 43.24
Best Fit 6 — χ2 = 43.32
Best Fit 7 — χ2 = 43.33
Best Fit 8 — χ2 = 43.37
Best Fit 9 — χ2 = 43.37
Best Fit 10 — χ2 = 43.39
JWST NIRSpec/PRISM (Espinoza et al. 2025)
Data − Best Fit 1 (ppm)
Wavelength (μm)